Exam-style induction problems

Practice questions · Day 12 · Wed 4 Nov · pack M9

5. Practice

From your course files

PQ 7.11 (practice questions). A 2.2 kW three-phase induction machine has the following ratings : 2.2 kW, 50 Hz, 1420 rpm, 415 V, 4.9 A, 0.80 efficiency, 0.78 power-factor.

(a) Calculate the synchronous speed in rpm
(b) Calculate the rated output torque in Nm
(c) Calculate the rated input power to the machine
  i) Using the voltage, current and power-factor
  ii) Using the output and efficiency
(d) Calculate the ideal efficiency at rated speed. Why is this different to the actual efficiency?

PQ 7.16 (practice questions). Constant volts-per-Hertz control is applied to an induction motor which is rated at 460 V and 60 Hz in order to keep the flux in the motor at its rated value. Calculate the required stator voltage corresponding to a stator frequencies of 30 Hz and 50 Hz.

PQ 7.12 (practice questions; do (c)–(d) here). The following tests are performed on a 415 V, 1455rpm, 50 Hz three-phase induction motor which is star-connected.

Stator resistance measurement : 3 ohms per phase
No-load test : input voltage 415 V, input current 3.0 A, input power 205 W
Blocked-rotor test : input voltage 65 V, input current 4.8 A, input power 350 W

(a) Find the simplified equivalent circuit parameters.
(b) Draw the simplified equivalent circuit showing the calculated parameter values.
(c) The motor is running at rated speed. Calculate the magnitude of the input current, the power-factor, the input power, output torque and output power, and hence the efficiency of the

motor.
(d) Calculate the rated copper losses, iron loss and total loss. Using the output power calculated in part (c), hence calculate the efficiency of the motor.

PQ 7.14 (practice questions; do (a)–(d) and (f) here). A 415 V, 50 Hz, 2790 rpm, 3 phase, star-connected induction motor has the following equivalent circuit parameters (all referred to the stator winding) : stator leakage reactance X_1 = 0.3 ohm, stator resistance R_1 = 0.2 ohms, rotor leakage reactance X_2' = 0.3 ohm, rotor resistance R_2' = 0.16 ohm and magnetising reactance X_M = 12 ohm. You can ignore the core-loss resistance. Use the simplified equivalent circuit for the following calculations. The motor is operated at rated voltage.

(a) A locked-rotor test (slip = 1) is performed at rated voltage. Ignoring the magnetising reactance for this part only, calculate the input current magnitude, input power and output torque
(b) At rated speed find the input current magnitude, input power factor and input power (do not neglect the magnetising reactance!).
(c) At rated speed find the rated output torque and output power and hence the efficiency.
(d) What are the starting torque and starting current as a fraction of the rated torque and rated current respectively?
(e) Determine the maximum torque available and the speed at which it occurs. What is the maximum torque as a fraction of rated torque?
(f) Calculate the magnitude of the input current at no-load. What is this as a fraction of rated current?
(g) Using the calculated torque at starting, maximum torque and rated torque, and your knowledge of the general shape of the induction motor torque versus speed characteristics, draw a rough sketch showing the output torque as a fraction of rated torque as a function of speed in rpm.

PQ 7.15 (practice questions; do (b)(i), (ii), (iv) here). A three-phase, 460 V, 60 Hz, 1760 rpm, wound-rotor induction motor has the following equivalent circuit parameters

R_1 = 0.25 ohm : R_2' = 0.2 ohm : X_1 = 1.2 ohm : X_2' = 1.1 ohm : X_M = 35 ohm

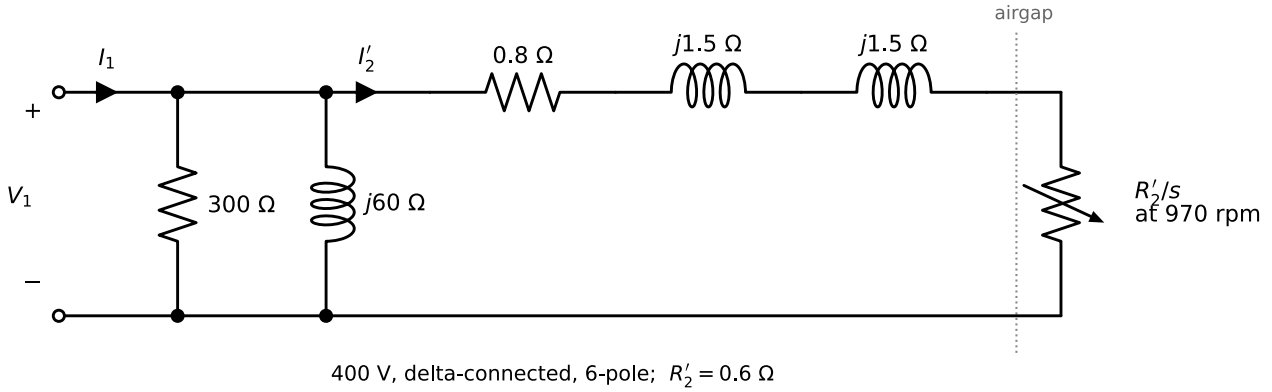
The motor is connected to a 3ph, 460V, 60Hz supply and is star-connected. Use the simplified equivalent circuit. Initially the external rotor resistance is set to zero.

(a) Calculate the speed (in rpm) at which maximum torque occurs.
(b) Using the torque equation, calculate the output torque at the following speeds (i) synchronous, (ii) rated speed, (iii) speed corresponding to maximum torque, (iv) locked rotor (0 rpm)
(c) Using the calculated torques in part (b), sketch out the torque versus speed characteristic.
(d) The total effective rotor resistance R_2' is adjusted by adding external rotor resistance. What is the total effective rotor resistance required to give maximum torque at standstill? What is the starting torque under this condition?

New questions

New Q1 (15 marks, easy–medium). A 400 V, 50 Hz, 6-pole, **delta-connected** induction motor runs at 970 rpm. Simplified-circuit parameters per phase: $R_1 = 0.8\ \Omega$, $R_2' = 0.6\ \Omega$, $X_1 = X_2' = 1.5\ \Omega$, $X_M = 60\ \Omega$, $R_C = 300\ \Omega$.

a. Find n_S, the slip, the rotor current frequency and R_2'/s. [3]
b. Find the phase current I_1 (magnitude and phase), the line current and the power factor. [5]
c. Find P_{IN}, P_{AG}, the output torque, the output power and the efficiency. [5]
d. Find the stator copper, rotor copper and iron losses, and show that they account for the difference between P_{IN} and P_{OUT}. [2]



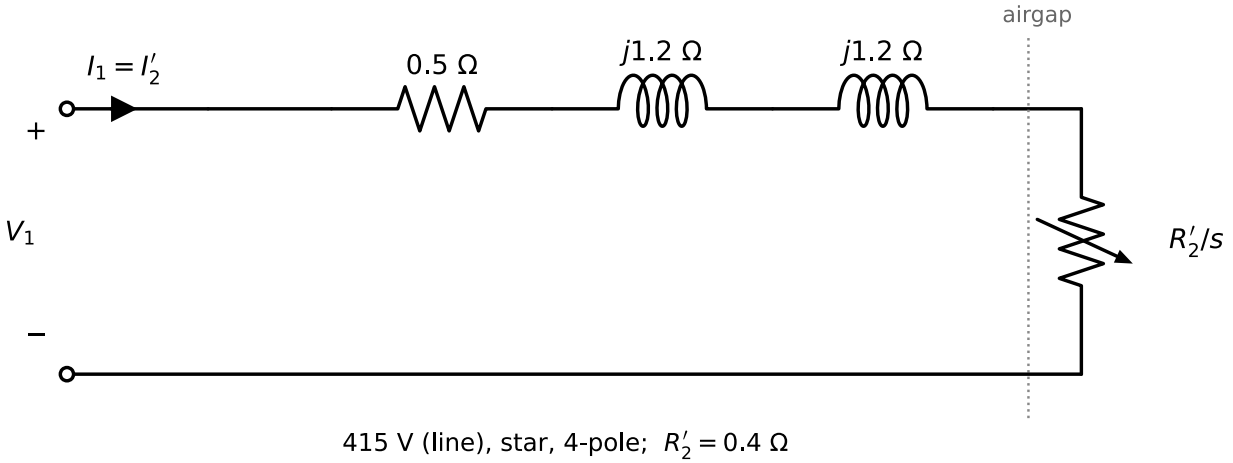
New Q1 circuit: 300 ohm parallel j60 ohm, then 0.8 ohm, j1.5 ohm, j1.5 ohm and R2′/s, delta phase voltage

New Q2 (17 marks, medium). A 415 V, 50 Hz, 4-pole, star-connected induction machine has $R_1 = 0.5\ \Omega$, $R_2' = 0.4\ \Omega$, $X_1 = X_2' = 1.5\ \Omega$, $X_M = 40\ \Omega$. Neglect R_C. It is driven by a wind turbine at 1530 rpm. Use the simplified equivalent circuit.

a. Is the machine motoring or generating? Find s and R_2'/s. [3]
b. Find I_2', I_1 (magnitude and phase), the real power and the reactive power drawn from the supply. Comment on the signs. [6]
c. Find P_{AG}, the shaft torque, the mechanical input power and the generator efficiency. [5]
d. Explain briefly why an induction generator connected to the grid cannot supply its own magnetising current, and what this means for the grid. [3]

New Q3 (18 marks, medium–hard). A 415 V, 50 Hz, 4-pole, star-connected induction motor has $R_1 = 0.5\ \Omega$, $R_2' = 0.4\ \Omega$, $X_1 = X_2' = 1.2\ \Omega$. Neglect X_M and R_C.

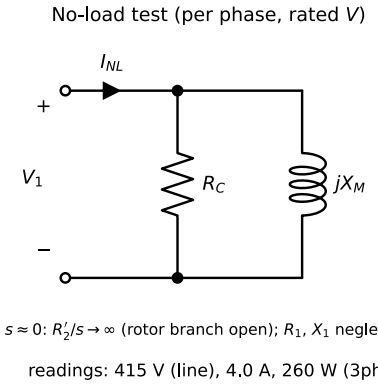
a. Find the torque and the input current magnitude at 1470 rpm. [3]
b. The motor drives a constant-torque load of 40 Nm. Find the exact operating slip and speed. Explain which root you choose. [6]
c. Estimate the speed in (b) using the linear torque–speed approximation and the result of (a). Comment on the accuracy. [3]
d. The supply voltage falls by 10% with the same 40 Nm load. Find the new speed (exact), and compare with the estimate $s \propto 1/V^2$. [4]
e. Find the starting torque. Can the motor start the 40 Nm load? Find the ratio of starting current to the running current in (b). [2]

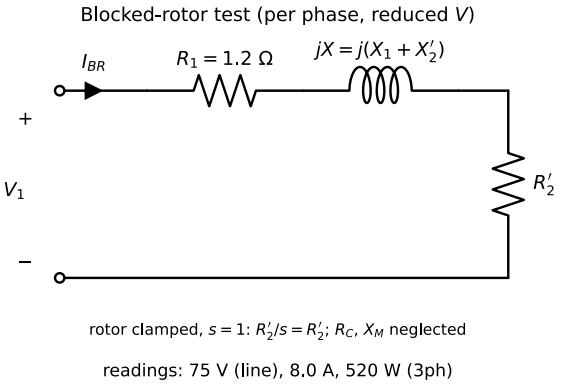


New Q3 circuit: 0.5 ohm, j1.2 ohm, j1.2 ohm and R2′/s, magnetising branch neglected

New Q4 (20 marks, hard: a full Q3-style chain). A 415 V, 50 Hz, 1440 rpm, star-connected induction motor gives these test results:

- DC stator resistance: 1.2 Ω per phase
- No-load test: 415 V, 4.0 A, 260 W
- Blocked-rotor test: 75 V, 8.0 A, 520 W





New Q4 test circuits: no-load 415 V, 4.0 A, 260 W; blocked rotor 75 V, 8.0 A, 520 W; R1 = 1.2 ohm

a. Find R_2', X, R_C and X_M for the simplified equivalent circuit. [4]
b. Find the number of poles, n_S and the rated slip. State the ideal efficiency at rated speed. [3]
c. At rated voltage and speed, find I_1 (magnitude and phase), pf, P_{IN}, τ, P_{OUT} and η. [7]

d. The same machine is driven at 1560 rpm on the same supply. Find the torque that must be applied, the electrical power delivered to the supply, and the generator efficiency. [4]
e. Explain why the actual efficiency in (c) is lower than the ideal efficiency in (b). [2]

6. Quick recall check

1. A 415 V star machine. What is V_1 in the per-phase circuit, and what is it for delta?
2. Which speed do you divide P_{AG} by to get torque, and which speed do you multiply τ by to get P_{OUT}?
3. In generator mode, which of s, R_2'/s, P_{IN}, Q_{IN}, τ are negative?
4. Write the rotor copper loss in terms of P_{AG}, and the mechanical power in terms of P_{AG}.
5. A quadratic for R_2'/s gives two roots. Which one is the normal running point?